UNIVERSITY OF TORONTO MISSISSAUGA
APRIL 2025 FINAL EXAMINATION
MAT102H5S - Introduction to Proofs
Multiple Choice Booklet

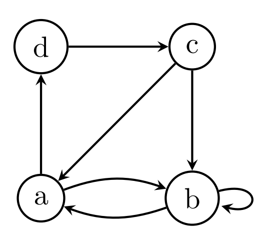
This booklet contains the multiple choice questions for **Question 7**. Each part of Question 7 includes five multiple choice answers, of which only a single response is correct. **All answers must be indicated on the scantron sheet (page 12) located at the end of the Exam Booklet.** *This* booklet will not be marked.

As no material in the booklet will be marked, you may remove the staple. The pages have been printed single sided, with the back side intended to be used for any rough work.

Table I.

P	Q	R	??
T	T	T	T
T	T	F	F
T	F	T	F
T	F	F	F
F	T	T	T
F	T	F	F
F	F	T	T
F	F	F	T

Diagram II.



Question 7: Each question is worth 1 mark. Indicate your answer by filling in the bubble sheet on the last page of the exam booklet. No part marks will be awarded for this question and you are not required to show your work.

MC 1. Consider Table I on the front page of this booklet. Which expression could be placed in the (??) to make the truth table valid?

A. $(P \vee Q) \implies (Q \wedge R)$

B. $(P \wedge Q) \implies (Q \vee R)$

C. $Q \implies (P \wedge R)$

D. $(R \vee Q) \implies P$

E. None of the above.

MC 2. Suppose that $A, B \subseteq \mathbb{Z}$. We say that

- B **smashes** A if $\forall a \in A, \exists b \in B, a \leq b$.
- B **crushes** A if $\forall a \in A, \forall b \in B, a \leq b$.
- B **dominates** A if $\exists a \in A, \forall b \in B, a \leq b$.
- B **obliterates** A if $\exists a \in A, \exists b \in B, a \leq b$.

If A is the set of even integers and B is the set of odd integers, which of the statements above describes the relationship between A and B?

A. B smashes and crushes A.

B. B crushes and dominates A.

C. B crushes and obliterates A.

D. B smashes and obliterates A.

E. None of the above.

MC 3. Suppose that a is an arbitrary positive integer such that $7 \mid a + 5$. Which of the following statements is **true**?

A. $7 \mid a$

B. $7 \mid (7a - 2)$

C. $7 \mid a^2$

D. $5 \mid a$

E. None of the above.

MC 4. When the Euclidean Algorithm is applied to the numbers 603 and 270, we get the following steps:

$$\begin{array}{ll} 603 = 270(2) + 63 & \quad 63 = 18(3) + 9 \\ 270 = 63(4) + 18 & \quad 18 = 9(2) + 0 \end{array}$$

If we reverse these steps to find a solution to $603 + 270 = 3$, what solution do we attain?

A. $x = 103$ and $y = -230$

B. $x = -77$ and $y = 172$

C. $x = 13$ and $y = -29$.

D. $x = 193$ and $y = -431$

E. None of the above.

MC 5. Consider Diagram II shown on the front page of this booklet. Define a relation $\sim$ on the set $A = \{a, b, c, d\}$ by saying that $x \sim y$ if there is an arrow from x to y. Which of the following statements is true?

A. R is symmetric, transitive, and symmetric.

B. R is reflexive and symmetric, but not transitive.

C. R is symmetric and transitive, but not reflexive.

D. R is symmetric, but not transitive or reflexive.

E. None of the above.

MC 6. How many solutions are there to the linear congruence relation $95x \equiv 20 \pmod{38}$, where $0 \leq x \leq 37$?

A. 0

B. 2

C. 3

D. 5

E. None of the above.

MC 7. Suppose that $f : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ is given by $f(x, y) = x^2 + y^2$. Which of the following is the preimage $f^{-1}((-\infty, 1))$?

A. $(0, 1) \times (0, 1)$

B. $\{(x, y) \in \mathbb{R} \times \mathbb{R} : 0 \le x^2 + y^2 < 1\}$.

C. $\left\{(x, y) \in \mathbb{R} \times \mathbb{R} : 0 \le x < 1, 0 \le y \le \sqrt{1 - x^2}\right\}$

D. $(-\infty, 1) \times (-\infty, 1)$

E. None of the above.

MC 8. Define the function $f : \mathbb{N} \to \mathcal{P}(\mathbb{N})$ by $f(n) = \{n, 2n, 3n, 4n, \ldots\}$. Which of the following statements is correct?

A. f is both injective and surjective.

B. f is surjective, but not injective.

C. f is injective but not surjective.

D. f is neither injective nor surjective.

E. f is not a well-defined function.

MC 9. Suppose that A and B are finite sets of the same cardinality. Which of the following statements is **true**?

A. There is an injective map $f : A \to B$ that is not surjective.

B. Every injective map $f : A \to B$ is also surjective.

C. Every function $f : A \to B$ is a bijection.

D. There are no bijective maps from A to B.

E. None of the above statements are true.

MC 10. Which of the following sets is **not** countably infinite?

A. $\{f : \mathbb{N} \to \{0, 1\}\}$

B. $\left\{\frac{a}{2^n} : a \in \mathbb{Z}, n \in \mathbb{N}\right\}$

C. $\left\{1 - \frac{1}{n} : n \in \mathbb{N}\right\} \cup \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$

D. $\{x \in (0, 1) : x^2 \in \mathbb{Q}\}$.

E. All of the above are countably infinite.